INTERNSHIP REPORT PRESENTATION

Development of Cybersecurity Utility Tools

Password Strength Evaluator • Vulnerability Scanner • Phishing Email Detection • Secure Login System

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Password Strength

Vulnerability Scan

Phishing Detection

Secure Login

OVERVIEW

Agenda

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3. Project Objectives
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Introduction

Cybersecurity has become one of the most critical concerns in the modern digital landscape. During this internship, four complementary cybersecurity utility tools were studied, designed and implemented — addressing different layers of information security:

- Strength of individual user credentials
- Security posture of a network or web application
- Human-targeted threat of phishing
- Technical safeguarding of the login process

This presentation documents the objectives, design, implementation and outcomes of each sub-project, along with the concepts of applied cryptography, network scanning, ML-based classification, and secure session management studied along the way.

Password Strength Evaluator

Vulnerability Scanner

Phishing Email Detection Model

Secure Login System

Organization Profile

Internship undertaken at Thiranex

[Brief description of organisation —Remote work]

Mentorship of Hariharan M

Internship duration: 4 weeks

Note: fields above are placeholders — fill in with your actual organisation and mentor details before presenting.

Problem Statement

Weak Passwords

Short, reused, or low-complexity passwords remain a leading cause of account compromise.

Unpatched Networks

Open ports, weak configurations, and outdated software often go unnoticed by administrators.

Phishing Attacks

Deceptive emails mimicking legitimate senders are hard to catch manually at scale.

Insecure Logins

Poor password storage and input validation expose systems to credential theft and injection.

Project Objectives (1/2)

Password Strength Evaluator

- Check length, character complexity, and uniqueness of passwords.
- Suggest stronger alternatives when a weak password is entered.
- Optionally check against a history of previously used passwords.

Vulnerability Scanner

- Scan a target host/network for open ports and weak configurations.
- Identify outdated software versions on discovered services.
- Generate a simple, readable vulnerability report.

Project Objectives (2/2)

Phishing Email Detection Model

- Train an ML model on labelled phishing / legitimate emails.
- Extract features: embedded URLs, sender patterns, keywords.
- Classify emails and report accuracy + confusion matrix.

Secure Login System

- Registration & login using salted hashing (bcrypt / Argon2).
- Input validation and parameterised queries to block SQL injection.
- Session management with logout; optional Two-Factor Authentication.

Scope of the Project

The scope of this internship was limited to building functional, educational-grade prototypes rather than production-ready security products:

Password Evaluator & Login System	Built as standalone / web-based demonstrations.
Vulnerability Scanner	Scoped to systems/networks with explicit permission (local/lab environments) — responsible, ethical testing only.
Phishing Detection Model	Scoped to publicly available, pre-labelled email datasets rather than live email traffic.

Together, the combined scope gives a rounded, hands-on introduction to authentication security, network reconnaissance, applied ML for security, and secure web development.

Technologies & Tools Used

Project	Technologies / Tools
Password Strength Evaluator	Python, regex, zxcvbn / custom scoring logic, SQLite (reuse-check database)
Vulnerability Scanner	Python, nmap / python-nmap, socket programming, banner grabbing
Phishing Email Detection Model	Python, Scikit-learn, Pandas, NumPy, NLTK, TF-IDF, Matplotlib/Seaborn
Secure Login System	Python (Flask/Django) or Node.js, bcrypt/Argon2, HTML-CSS-JS, SQLite/MySQL, TOTP-based 2FA

System Architecture

Password Evaluator

Input → Scoring Engine → Strength Score & Feedback → History Lookup → Suggested Password

Vulnerability Scanner

Target Input → Port Scan → Service/Version Fingerprint → Vulnerability Match → Report

Phishing Detection

Email Dataset → Preprocessing → Feature Extraction (URL/TF-IDF) → Classifier → Phishing / Safe

Secure Login

Form → Input Validation → Password Hashing → Database → Session Manager → 2FA → Dashboard

Methodology

1

Requirement Study

Reviewed password policies, OWASP guidelines, and existing datasets.

2

Design

Defined architecture and data flow for each of the four modules.

3

Implementation

Built the evaluator, scanner logic, ML model, and login system.

4

Testing

Validated against sample passwords, a lab network, a test split, and session checks.

5

Documentation

Recorded findings, screenshots, accuracy metrics and reports.

Expected Outcomes

Password Evaluator

Working understanding of password security & basic cryptography via reliable strength scoring.

Vulnerability Scanner

Practical insight into penetration testing — open ports, weak configs, and outdated software surfaced clearly.

Phishing Detection

High-accuracy Phishing / Safe classification based on textual content and URL features.

Secure Login System

Hashed passwords, validation, sessions & optional 2FA — significantly reducing unauthorised access.

Bibliography / References

- OWASP Foundation — OWASP Testing Guide & OWASP Top Ten (owasp.org)
- NIST SP 800-63B — Digital Identity Guidelines: Authentication & Lifecycle Management
- Scikit-learn Developers — Scikit-learn: Machine Learning in Python (scikit-learn.org)
- Python Software Foundation — Python 3 Documentation (docs.python.org/3)
- bcrypt / Argon2 project documentation — password hashing algorithm reference
- Nmap Project — Nmap Network Scanning: Official Documentation (nmap.org/book)

Verify each citation/link before final submission — exact editions or URLs may need updating.

Thank You

Questions & Discussion

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